

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA
MID SEMESTER EXAMINATION
SESSION: 2024 – 2025 (SPRING)

Subject code: CS6218

Subject Name: Machine Learning

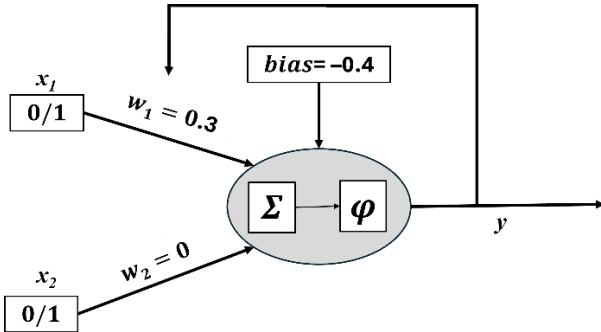
Dept. Code: CS

No. of pages: 2

Full Marks: 30

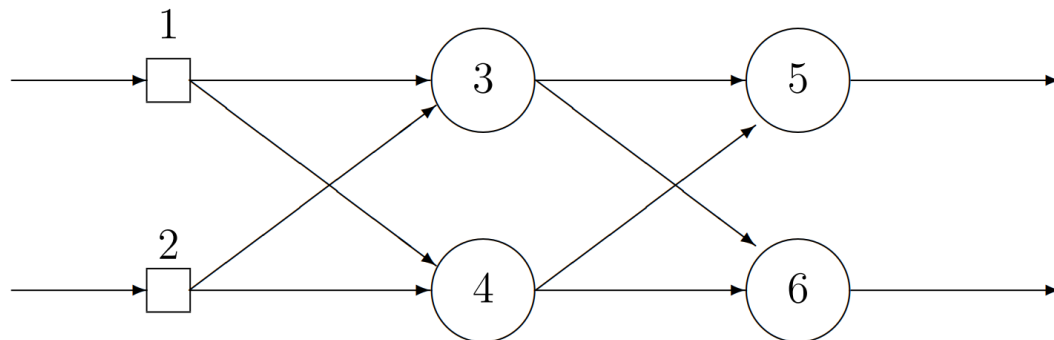
Duration: 2 Hrs

Instructions: Answer ALL questions. Each question carries 6 marks.

Question	CO & BL												
1. Explain with suitable example, why feature scaling is important? Consider the data [200, 300, 400, 600, 1000]. Normalize these data with Max normalization, Mean normalization, and Z-score normalization.	CO1 BL:1&3												
2. Given a linearly separable dataset and that the perceptron uses a transfer function as follows: $\varphi(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise.} \end{cases}$ Present an analysis of why the perceptron rule of updating the weights will always converge towards successful weight values, for some properly chosen learning rate.  Consider a perceptron to represent Boolean AND function with two inputs (x_1 and x_2) with initial weights and bias values as shown above. Following the perceptron update rule, show the update of weights in each epoch for each input data, till convergence.	CO2 BL: 3&4												
3. The dataset of pass/fail in an exam for 5 students is given below. <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Hours study (x)</th><th>Pass(1) or Fail(0) (y)</th></tr> </thead> <tbody> <tr><td>29</td><td>0</td></tr> <tr><td>15</td><td>0</td></tr> <tr><td>33</td><td>1</td></tr> <tr><td>28</td><td>1</td></tr> <tr><td>39</td><td>1</td></tr> </tbody> </table> If we use logistic regression as a classifier and assume that the input to the logistic function is modelled as $z = 2x - 64$ then - Calculate the probability of pass for a student who studies 33 hours. - How many hours a student should study to ensure probability of passing at least 95%?	Hours study (x)	Pass(1) or Fail(0) (y)	29	0	15	0	33	1	28	1	39	1	CO2 BL:2&3
Hours study (x)	Pass(1) or Fail(0) (y)												
29	0												
15	0												
33	1												
28	1												
39	1												
4. Consider a linear regression problem with one variable w . If mean squared error is to be used as cost function $J(w)$, then show pictorially how $J(w)$ varies with w . Irrespective of random initialization of w , how it is guaranteed that the gradient descent algorithm will converge with a proper choice of learning rate α . Also explain	CO3 BL:1&2												

the impact of α being too small or too large. You may use drawings to support your explanation.

5. The following diagram represents a feed-forward neural network with one hidden layer:



A weight on connection between nodes i and j is denoted by w_{ji} , such as w_{31} is the weight on the connection between nodes 1 and 3. The weights in the network are as given below: $w_{31}=-2$, $w_{32}=3$, $w_{41}=4$, $w_{42}=-1$, $w_{53}=1$, $w_{54}=-1$, $w_{63}=-1$, $w_{64}=1$.

Each neuron uses step activation function. Each of the inputs (1 and 2) are binary. Calculate the output of the network at 5 and 6 for each of the following input patterns.

Node 1	0	1	0	1
Node 2	0	0	1	1

CO3
BL:3&4